Week 4: AI screens and drafts; engineers keep shutdown authority

Weekly evidence artifact · Week 4 · Trimester 1, 2026

1. Theory anchor

Human–AI task allocation matrix [10].

2. Artifact

Tier	Tasks
Human only	Emergency block shutdowns, licensing, Subsoil compliance, contract renegotiation
Human + AI	Reserve-decline modelling, drill patterns, inventory swaps
AI + human review	Acid-dosing alerts and draft valve orders (Lead Engineer signs off any shift above ±5%)
Automated	SCADA telemetry filtering, robotic drum filling at Ulba

3. Company A vs. Company B insight

Cameco keeps people in control of remote systems [3]; the same gate idea suits Kazatomprom’s 20,000+ wellheads.

4. Transfer verdict

Pilot the ±5% review gate.

5. Evidence + verification

Tiers based on [6] [7] [10]; thresholds are my proposals.

6. Mini-recommendation

The Lead Engineer should approve every dosing shift above ±5%; progress signal: gate log with zero unsigned shifts.

7. Learning progression & AI disclosure

I used to think automation meant fewer people. Now I design who decides. AI suggested the four-tier format; I set the tasks and limits.

Companies: Cameco (A) and NAC Kazatomprom JSC (B). Source numbers [n] refer to the Evidence page of the ePortfolio.
Proposed targets and thresholds are design assumptions and pilot targets, not verified Kazatomprom data.